

Taehei Kim

hayleyy321@gmail.com · hayleymol.github.io
Google Scholar · Daejeon, South Korea

Research interests Mixed reality telepresence; spatial computing; scene synthesis and generation; human-centered XR systems.

Professional Experience

- 2026.09– **Postdoctoral Researcher**, InnoCORE AI-Meta Scientist Program, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea.
Host: Prof. Tae-Kyun Kim.
- 2024.09– **Co-Founder**, BerryHi, Daejeon, South Korea.
Mixed reality telepresence systems; shipped *Parameter Room* on the Meta Quest Store.

Education

- 2021.03–2026.08 **Ph.D., Culture Technology**, KAIST, Daejeon, South Korea.
Advisor: Prof. Sung-Hee Lee.
Dissertation: *Dynamic Space Optimization and Adjoining for Transmitting User and Spatial Context in Mixed Reality Telepresence*.
- 2018.08–2021.02 **M.S., Culture Technology**, KAIST, Daejeon, South Korea.
Advisor: Prof. Sung-Hee Lee.
Thesis: *Quadruped Locomotion on Non-Rigid Terrain using Reinforcement Learning*.
- 2013.03–2018.08 **B.A. in Asian Studies** and **B.S. in Computer Science** (dual degree), Yonsei University, Seoul, South Korea.

Honors and Awards

- 2026.07 Best Poster Award, Korea Computer Graphics Society (KCGS).
- 2025.11 Excellence Award, Metaverse Developer Contest, Korea Radio Promotion Association (RAPA).
- 2024.10 Best Demo Honorable Mention, IEEE ISMAR.
- 2021.03–2026.08 Fully Funded Ph.D. Fellowship, KAIST.
- 2021.09 Tenacity Scholarship, KAIST.
- 2021.08 First Place, “Hello, World!: New Playground”, Art Center NABI.
- 2018.08–2021.02 Fully Funded M.S. Fellowship, KAIST.

Publications

†first author *equal contribution

Peer-Reviewed Journal Articles

- [J4] **Taehei Kim**[†], J. Shin, H. Kim, H. Jang, J. Kang, and S.-H. Lee. “Voronoi Rooms: Dynamic Visibility Modulation of Overlapping Spaces for Telepresence.” *ACM Transactions on Graphics (TOG)*, 45(2), 2026.
[doi:10.1145/3777900](https://doi.org/10.1145/3777900). Presented at ACM SIGGRAPH 2026.
- [J3] J. Kang, **Taehei Kim**, H. Kim, and S.-H. Lee. “Real-Time Translation of Upper-Body Gestures to Virtual Avatars in Dissimilar Telepresence Environments.” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 31(10):8711–8725, 2025. [doi:10.1109/TVCG.2025.3577156](https://doi.org/10.1109/TVCG.2025.3577156).
- [J2] **Taehei Kim**[†], H. Kim, J. Lee, and S.-H. Lee. “Evaluating User Perception toward Physics-Adapted Avatars in Remote Heterogeneous Spaces.” *Computers & Graphics*, 128:104185, 2025. [doi: 10.1016/j.cag.2025.104185](https://doi.org/10.1016/j.cag.2025.104185)
- [J1] D. Yang, J. Kang, **Taehei Kim**, and S.-H. Lee. “Visual Guidance for User Placement in Avatar-Mediated Telepresence between Dissimilar Spaces.” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 30(12):7558–7570, 2024. [doi: 10.1109/TVCG.2024.3354256](https://doi.org/10.1109/TVCG.2024.3354256)

Peer-Reviewed Full Conference Papers

- [C3] **Taehei Kim**[†], H. Kim, J. Shin, H. Kim, and S.-H. Lee. “Neighborhood: Dynamic Spatial Alignment for Traversable Telepresence across Remote Spaces.” *ACM Symposium on User Interface Software and Technology (UIST)*, 2026. doi:10.1145/3830398.3830524
- [C2] H. Kim, **Taehei Kim**^{*}, J. Shin^{*}, H. Kim, and S.-H. Lee. “TranSpace: Progressive Anchoring for Metric-Consistent Scene Synthesis.” *ACM International Conference on Multimedia (MM)*, 2026. doi:10.1145/3767308.3836316.
- [C1] J. Kang, D. Yang, **Taehei Kim**, Y. Lee, and S.-H. Lee. “Real-Time Retargeting of Deictic Motion to Virtual Avatars for Augmented Reality Telepresence.” *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 885–893, 2023. doi: 10.1109/ISMAR59233.2023.00104

Peer-Reviewed Short Papers, Posters, and Demonstrations

- [D4] J. Shin, H. Kim, E. Lee, D. Shin, K. Lee, **Taehei Kim**, H. Kim, J. An, and S.-H. Lee. “Situating Embodied XR Agents via Spatial Reasoning and Prompting.” *IEEE ISMAR Adjunct*, pp. 933–934, 2025. doi:10.1109/ISMAR-Adjunct68609.2025.00255.
- [D3] **Taehei Kim**[†], J. Shin, H. Kim, H. Jang, J. Kang, and S.-H. Lee. “Visibility Modulation of Aligned Spaces for Multi-User Telepresence.” *IEEE ISMAR Adjunct*, pp. 626–627, 2024. **Best Demo Honorable Mention.**
- [D2] H. Jang^{*}, **Taehei Kim**^{*}, S. Oh, J. Lee, S. Lee, and S. H. Yoon. “Sense of Embodiment Inducement for People with Reduced Lower-Body Mobility and Sensations with Partial-Visuomotor Stimulation.” *ACM SIGGRAPH Emerging Technologies*, pp. 1–2, 2022.
- [D1] H.-I. Kim, **Taehei Kim**, E. Song, S. Y. Oh, D. Kim, and W. Woo. “Multi-Scale Mixed Reality Collaboration for Digital Twin.” *IEEE ISMAR Adjunct*, pp. 435–436, 2021.

Preprints and Manuscripts under Review

- [P2] One paper currently under review, 2026.
- [P1] **Taehei Kim**[†] and S.-H. Lee. “Quadruped Locomotion on Non-Rigid Terrain using Reinforcement Learning.” *arXiv:2107.02955*, 2021.

Grants and Funded Research Projects

2026.06–	KEP Startup Fund — <i>Real-Space Overlay Scene Generation for VR Crowdsourcing Data Collection</i> . USD 37,000. Role: Internal director.
2024.05–2025.12	Korea Radio Promotion Association (RAPA) — <i>Mixed Reality Telepresence System</i> . Role: Student project lead).
2024.08–2024.09	Daejeon Art Center Joint New Technology Performance. USD 11,000. Role: Lead director.
2022.03–2024.02	National Research Foundation of Korea — <i>Mixed Reality Scenes in Heterogeneous Spaces</i> . Role: Student project lead.
2021.03–2022.02	Korea Institute for Advancement of Technology (KIAT) — VR/AR Expert Training Program. Role: Participating researcher.

Public Demonstrations, Exhibitions, and Deployed Systems

2025.11	<i>Parameter Room</i> , Mixed reality telepresence application, Meta Quest Store (BerryHi).
2024.10	<i>Visibility Modulation of Aligned Spaces for Multi-User Telepresence</i> , IEEE ISMAR 2024 Demonstrations, Seattle, USA. Best Demo Honorable Mention.
2024.09	X-Space, Daejeon Art Center, Daejeon, South Korea.
2022.07	<i>Sense of Embodiment Inducement with Partial-Visuomotor Stimulation</i> , ACM SIGGRAPH 2022 Emerging Technologies, Vancouver, Canada.

Teaching Experience

Guest Lectures and Instruction

- 2021.08–2023.07 Metaverse Instructor. Designed and delivered XR content-creation curricula for public and corporate audiences. Clients: Dangjin Cultural Foundation, Daejeon Culture and Arts Foundation, Hanwha Galleria, 3PROTV.

Courses Prepared to Teach

Computer Graphics; Virtual and Augmented Reality; Human-Computer Interaction; Generative Models and Agents for 3D Content Generation.

Mentoring

2024.03- Mentored M.S. students (Hyeslim Kim, Jihun Shin, Hyeonjin Kim) at KAIST LAVA Lab and in collaborating laboratories (Taeyeon Kim). Mentorship resulted in co-authored publications [J4], [J2], [C3], [C2], [D4], [D3].

Technical Skills

Programming C#, Python, C++.

XR development Unity, Mixed Reality Toolkit (MRTK), Photon networking, Meta Quest 3, HTC Vive.

Methods Computational geometry and optimization, LLM/diffusion-based scene synthesis, quantitative and qualitative HCI evaluation.

Languages

Korean (native); English (fluent).

References

Sung-Hee Lee, Professor, Graduate School of Culture Technology, KAIST, sunghee.lee@kaist.ac.kr (Ph.D. advisor).

Tae-Kyun Kim, Professor, School of Computing, KAIST, kimtaekyun@kaist.ac.kr (Postdoctoral host).